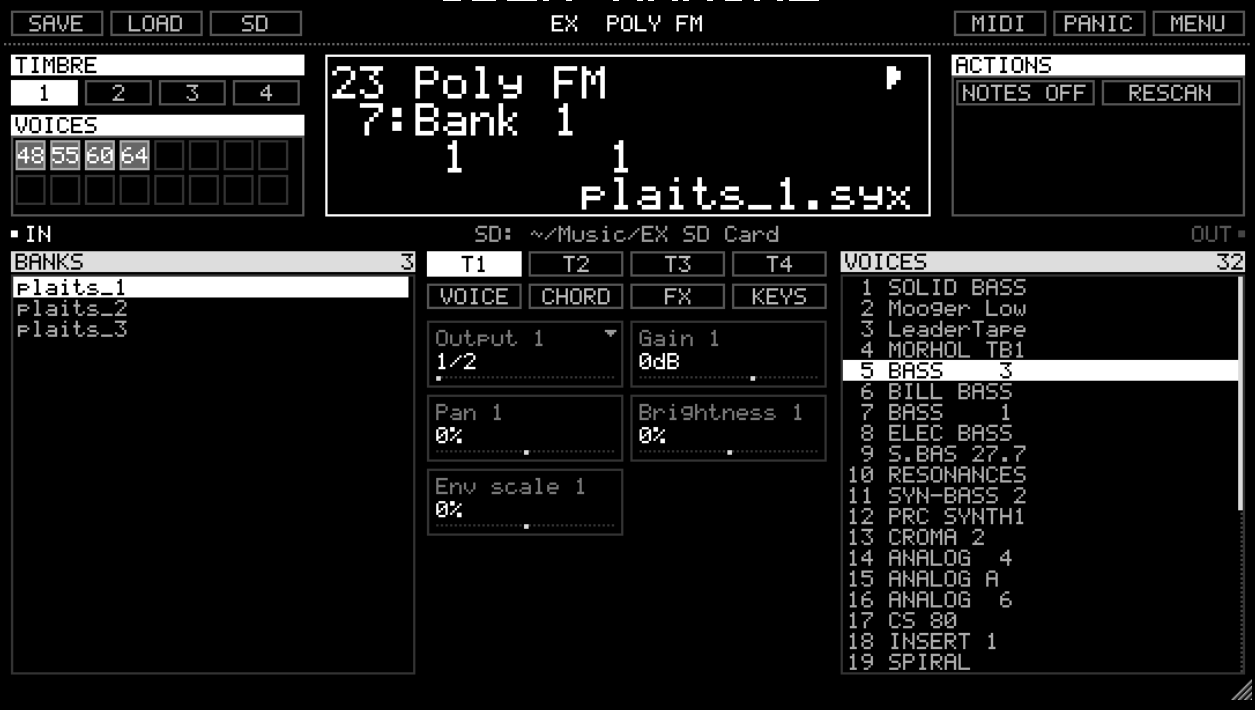
EX-GF

USER MANUAL



Four disting EX-style instruments: **EX Multisample** · **EX Poly Wavetable** · **EX Poly FM** · **EX Poly Exciter**

VST3 · AU · LV2 · standalone - macOS, Windows and Linux

Version 1.0.2 - github.com/femifleming/ex-gf

CONTENTS

INTRODUCTION 3
INSTALLING 3
THE SCREEN 5
THE SD CARD FOLDER 7
EX MULTISAMPLE 8
EX POLY WAVETABLE 15
EX POLY FM 18
EX POLY EXCITER 21
TROUBLESHOOTING 24
CREDITS AND LICENCE 24

INTRODUCTION

EX-GF is a suite of four software instruments modelled on algorithms of the Expert Sleepers **disting EX** Eurorack module. They follow the module's manual (firmware 1.26) parameter for parameter, use the same file formats (sample folders, wavetables, exciters, DX7 banks, Scala tunings) and wear the module's look: the whole window is one white-on-black pixel display in the module's own 5×7 font.

Plug-in	Algorithm(s)	In short
EX Multisample	3 - SD Multisample 5 - WAV Recorder	16-voice, 3-timbre sample player with velocity layers, round robins, chords, arpeggiator, EQ and delay; plus a recorder with an **auto-sampler** that samples any MIDI/CV instrument into a ready-to-play folder.
EX Poly Wavetable	10 - Poly Wavetable	16-voice wavetable synth: 2 envelopes, filter, LFO, unison, Juno-style chorus and delay. Imports WaveEdit banks.
EX Poly FM	23 - Poly FM	16-voice, 4-timbre 6-operator FM (the Plaits engine). Loads DX7 .syx banks; DX7 editors can edit it over SysEx.
EX Poly Exciter	13 - Poly Exciter	16-voice exciter/resonator synth: plucked, struck and bowed-like sounds from short exciter samples or live input.

Each instrument is available as a **VST3** (all platforms), **Audio Unit** (macOS), **LV2** (Linux) and a **standalone app**. Settings are saved with your DAW project, and every plug-in also remembers its last state between sessions.

EX-GF is not affiliated with or endorsed by Expert Sleepers; "disting" is their trademark and is used only to describe compatibility. If you own the module, its MicroSD card works directly with these plug-ins.

INSTALLING

All downloads are on the release page: **github.com/femifleming/ex-gf/releases**.

MACOS - HOMEBREW

```
brew tap femifleming/ex-gf https://github.com/femifleming/ex-gf
brew trust femifleming/ex-gf
brew install --cask ex-gf            # apps + VST3 + AU
brew install --cask ex-gf-sd-card    # optional sample library
```

Homebrew 7 asks you to *trust* third-party taps once. To build from source instead (macOS or Linux): `brew install ex-gf`, then follow the link instructions it prints.

MACOS - DOWNLOAD

- **EX-GF-1.0.2-macOS.zip**: unzip and double-click **install.command** - apps go to ~/Applications, plug-ins to ~/Library/Audio/Plug-Ins/VST3 and /Components.
- **EX-GF-1.0.2-macOS.pkg**: installs for all users (right-click -> Open the first time).
- The builds are universal (Apple silicon + Intel) but not notarised. If macOS or your DAW refuses to open them, run:

```
xattr -dr com.apple.quarantine ~/Library/Audio/Plug-Ins/VST3/EX*.vst3 \
    ~/Library/Audio/Plug-Ins/Components/EX*.component ~/Applications/EX*.app
```

WINDOWS

Unzip **EX-GF-1.0.2-Windows-x64.zip**, right-click **install.ps1** -> *Run with PowerShell* (as administrator). VST3 plug-ins go to C:\Program Files\Common Files\VST3; the apps appear in the Start menu under **EX-GF**.

LINUX

```
tar xzf EX-GF-1.0.2-Linux-x86_64.tar.gz
cd ex-gf-1.0.2-Linux-x86_64 && ./install.sh
```

VST3 -> ~/.vst3, LV2 -> ~/.lv2, apps -> ~/.local/bin with desktop menu entries.

THE SAMPLE LIBRARY

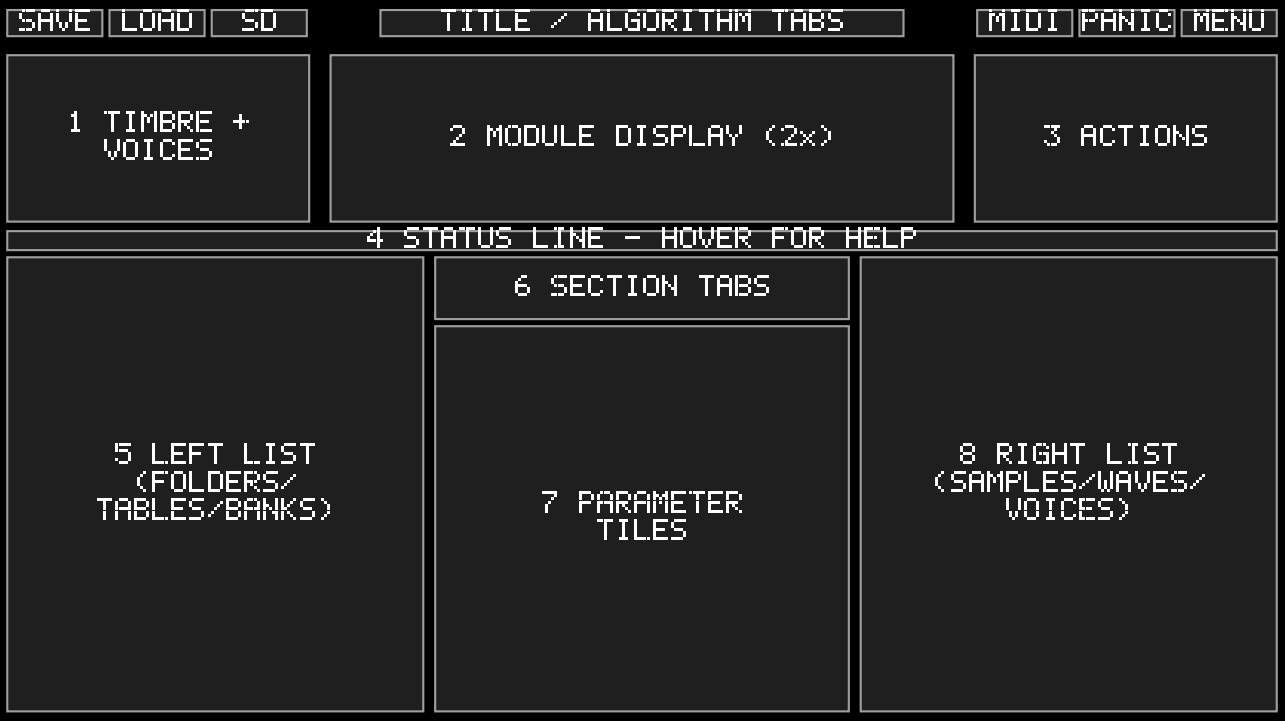
Download **EX-SD-Card.zip** and unzip it into your **Music** folder so you have **Music/EX SD Card** (the default location), or pick any folder with the **SD** button. It holds 18 multisampled instruments, 38 wavetables, 5 exciter tables, DX7 banks and Scala tunings - all free (CC0 / CC-BY / MIT, see CREDITS.txt inside). A real disting EX MicroSD card works too.

BUILDING FROM SOURCE

```
git clone --recursive https://github.com/femifleming/ex-gf
cd ex-gf
cmake -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build --config Release
```

THE SCREEN

Every plug-in uses the same mirror-symmetrical layout. The window can be resized; it snaps to whole-pixel sizes (1×, 2×, 3× …) so the pixels stay sharp.



	Area	What it does
	Top bar	**SAVE / LOAD** presets (stored on the SD card), **SD** chooses the card folder, **MIDI** channels, **PANIC** (all notes off), **MENU** (every option of the module's menus). In EX Multisample the centre holds the algorithm tabs.
1	Timbre / voices	Timbre buttons (multi-timbral plug-ins) choose which timbre the lists edit. The 16 cells show each voice's note and level.
2	Module display	The disting EX's own 128×32 screen: current parameter, live voices, levels, auto-sampler progress. Mouse wheel = parameter, shift+wheel = value; Up/Down and Left/Right act like the P and V encoders, Enter = push P (menu), Esc = push V.
3	Actions	Buttons for the algorithm: RECORD, PLAY, AUTO, TEST, IMPORT, WAVEEDIT, NOTES OFF, RESCAN …
4	Status line	Hover anything for a description. MIDI IN / OUT activity lights sit at either end.
5/8	Lists	Click to load: sample folders and their samples, wavetables and their waves, exciter tables and exciters, FM banks and voices, recordings.
6	Section tabs	Pages of related parameters (e.g. ENV, FILTER, CHORD, FX).
7	Parameter tiles	Drag up/down or use the wheel to change (hold shift for fine steps); double-click resets to the default. Tiles with a small (v) open a list of choices when clicked.

PLAYING NOTES

Play from MIDI, or click the window and use the computer keyboard: **A W S E D F T G Y H U J K O L P ;** play a chromatic octave and a half, **Z**/**X** shift the octave.

MIDI

- The receive channel defaults to 1 (Omni is available under MIDI). Multi-timbral plug-ins listen on consecutive channels (timbre 1 on the receive channel, timbre 2 on the next …) unless you assign each timbre its own channel.
- Recognised: note on/off, velocity, pitch bend, sustain (CC64), All Notes Off (CC123); channel and poly aftertouch in Poly Exciter; DX7 SysEx in Poly FM. *Delay uses MIDI clk* (menu) locks delay times to the host tempo.

OUTPUTS

Out 1/2 is the main stereo output. **Out 3/4** is a second stereo bus (enable it in your DAW) carrying per-timbre or per-voice outputs depending on the *Output mode*, the paraphonic gate and mono mix (as on the module), and the auto-sampler's CV/gate.

SAVING AND REMEMBERING

- In a DAW, each instance is saved with the project.
- **SAVE** writes a preset to *SD card/presets/<plug-in>*; **LOAD** lists them (plus *Reset to defaults*).
- Every plug-in and app also remembers its **last state**: it is written whenever you change something and when the plug-in closes, and a new instance starts from it. The files live in ~/Library/Application Support/EX-GF (macOS), %APPDATA%\EX-GF (Windows) or ~/.config/EX-GF (Linux).

THE SD CARD FOLDER

The plug-ins read a folder laid out exactly like the disting EX's MicroSD card (default **Music/EX SD Card**).

Folder	Contents
<any name>/	A **multisample folder** of WAV files (see naming below). Every top-level folder that contains WAVs is listed.
wavetables/	Each wavetable is a sub-folder of single-cycle WAVs (used alphabetically), or one WAV of concatenated cycles (600 samples each; 256 for WaveEdit banks - detected automatically). Optional *playlist-wavetable.txt* with -wavelength=.
exciters/	Each exciter table is a folder of up to 128 short WAVs, or one WAV containing exactly 128 equal-length exciters.
FMSYX/	DX7 32-voice bank SysEx files (.syx, 4104 bytes). Without any, three built-in banks are used.
scl/ kbm/	Scala scales and keyboard maps for the Scala/MTS and Scala KBM parameters.
!RECORD/	WAV Recorder recordings (REC00001.WAV …).
AUTO0001/ …	Auto-sampler output - ordinary multisample folders.
presets/	Presets saved with SAVE.

NAMING MULTISAMPLES

Token	Example	Meaning
_<note>	piano_A2.wav, piano_C#5.wav	Natural pitch. Sharps only, C4 = MIDI note 60.
_SW<n>	piano_A2_SW40.wav	Switch point: the lowest MIDI note that uses this sample.
_V<n>	snare_V1.wav, snare_V2.wav	Velocity layer; V1 is the softest.
_RR<n>	snare_RR1.wav, snare_RR2.wav	Round-robin variants of the same sample.
combined	cello_C3_V2_RR1.wav	Velocity first, then round robin.

- Files without a note name are mapped chromatically from note 48 (C3) upwards - handy for one-shots.
- Without explicit switch points, gaps of up to 3 semitones are covered by pitching the upper sample down; for larger gaps the lower sample is also pitched up over part of the gap (exactly as on the module).
- Loops are read from the WAV's *smpl* or *cue* chunks. A sample loops if it has loop points, otherwise it plays once.
- An optional *playlist.txt* ("disting playlist v1") can set -natural=, -switch=, -loop= and -endLoopOnRelease= per file.

EX MULTISAMPLE



3 - SD MULTISAMPLE

A 16-voice, 3-timbre sample instrument. Choose a folder in the left list; the right list shows its samples - click one to hear it. The display's top line shows the file, natural note and switch point of the highest playing voice.

- **Timbres**: the TIMBRE buttons pick which timbre the folder list sets. Timbres 2 and 3 can follow timbre 1 or use their own folders, and each has its own note range and arpeggiator.
- **SOUND**: gain, saturation, voice count, round-robin mode, output mode and stereo spread.
- **ENV**: attack/decay/sustain/release; release 127 = infinite (the sample plays to its end or loops forever).
- **TUNE**: octave, transpose, fine tune, bend range, Scala tuning and per-voice detune (an "old polysynth" spread).
- **CHORD**: chord generator (key, scale, shape, inversion), arpeggiator per timbre and broken chords (strum time/direction).
- **FX**: stereo or ping-pong delay and bass/treble EQ. **KEYS**: sustain behaviour, velocity curve and note ranges.
- MENU -> SD Multisample -> Options: MIDI note range, paraphonic gate mode, *Delay uses MIDI clk*, *Transpose limit*.

CHORDS AND ARPEGGIOS

Scales	Shapes	Arpeggio modes
Major, Natural Minor, Dominant, Fully Diminished, Dominant Dim, Augmented, Whole Tone, Chromatic, Harmonic Minor	None, Octave, Two Octaves, Root/Fifth, Root/Fifth + 8ve, Triad, Triad + 8ve, Sus4, Sus4 + 8ve, Sixth, Sixth + 8ve, Seventh, Seventh + 8ve, Ninth	Off, Up, Down, Alt, Alt2, Up/Down/Alt/Alt2 -8ve (skip the repeated octave), Step Up, Step Down, Random, Random2 (no repeats), Random3 (permutations), As Played

Each new note (gate) advances the arpeggio by one step through the chord built on it; with chords off, the arpeggio runs over the notes you are holding. *Range* 2 or 3 adds copies one or two octaves up.

PARAMETERS

#	Parameter	Range	Default	Page	Values / notes
1	Attenuverter 1	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
2	Attenuverter 2	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
3	Attenuverter 3	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
4	Attenuverter 4	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
5	Attenuverter 5	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
6	Attenuverter 6	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
7	Folder	0 .. 999	0	list	
8	Attack time	0 .. 127	0	ENV	
9	Decay time	0 .. 127	60	ENV	
10	Sustain level	0 .. 127	127	ENV	
11	Release time	0 .. 127	77	ENV	
12	Octave	-8 .. 8	0	TUNE	
13	Transpose	-60 .. 60	0	TUNE	
14	Fine tune	-100 .. 100 cents	0 cents	TUNE	
15	Gain	-40 .. 24 dB	0 dB	SOUND	
16	Saturation	0 .. 1	On	SOUND	0=Off, 1=On
17	Sustain	0 .. 1	Off	KEYS	0=Off, 1=On
18	Max voices	1 .. 16	16	SOUND	
19	Bend range	0 .. 48	2	TUNE	
20	Pitch bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
21	Voice 1 detune	-100 .. 100 cents	0 cents	TUNE	
22	Voice 2 detune	-100 .. 100 cents	0 cents	TUNE	
23	Voice 3 detune	-100 .. 100 cents	0 cents	TUNE	
24	Voice 4 detune	-100 .. 100 cents	0 cents	TUNE	
25	Voice 5 detune	-100 .. 100 cents	0 cents	TUNE	
26	Voice 6 detune	-100 .. 100 cents	0 cents	TUNE	
27	Voice 7 detune	-100 .. 100 cents	0 cents	TUNE	
28	Voice 8 detune	-100 .. 100 cents	0 cents	TUNE	
29	Chord enable	0 .. 7	Off	CHORD	0=Off, 1=On, 2=Gate 1, 3=Gate 2, 4=Gate 3, 5=Gates 1+2, 6=Gates 1+3, 7=Gates 2+3
30	Chord key	-12 .. 12	0	CHORD	
31	Chord scale	0 .. 8	Major	CHORD	0=Major, 1=Natural Minor, 2=Dominant, 3=Fully Diminished, 4=Dominant Dim, 5=Augmented, 6=Whole Tone, 7=Chromatic, 8=Harmonic Minor
32	Chord shape	0 .. 13	None	CHORD	0=None, 1=Octave, 2=Two Octaves, 3=Root/Fifth, 4=Root/Fifth + 8ve, 5=Triad, 6=Triad + 8ve, 7=Sus4, 8=Sus4 + 8ve, 9=Sixth, 10=Sixth + 8ve, 11=Seventh, 12=Seventh + 8ve, 13=Ninth
33	Chord inversion	0 .. 3	0	CHORD	
34	Arpeggio 1 mode	0 .. 14	Off	CHORD	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
35	Arpeggio 2 mode	0 .. 14	Off	CHORD	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played

#	Parameter	Range	Default	Page	Values / notes
36	Arpeggio 3 mode	0 .. 14	Off	CHORD	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
37	Arpeggio 1 range	1 .. 3	1	CHORD	
38	Arpeggio 2 range	1 .. 3	1	CHORD	
39	Arpeggio 3 range	1 .. 3	1	CHORD	
40	Scala/MTS	-16 .. 99	0	TUNE	
41	Scala KBM	0 .. 99	0	TUNE	
42	Folder 2	-1 .. 999	-1	list	
43	Folder 3	-1 .. 999	-1	list	
44	Min note 1	0 .. 127	0	KEYS	
45	Max note 1	0 .. 127	127	KEYS	
46	Min note 2	0 .. 127	0	KEYS	
47	Max note 2	0 .. 127	127	KEYS	
48	Min note 3	0 .. 127	0	KEYS	
49	Max note 3	0 .. 127	127	KEYS	
50	Output spread	-100 .. 100 %	0 %	SOUND	
51	Delay mode	0 .. 2	Off	FX	0=Off, 1=Stereo, 2=Ping-pong
52	Delay level	-40 .. 0 dB	-3 dB	FX	
53	Delay time	1 .. 1365 ms	500 ms	FX	
54	Delay feedback	0 .. 100 %	50 %	FX	
55	Tone bass	-240 .. 240	0	FX	
56	Tone treble	-240 .. 240	0	FX	
57	Break time	0 .. 1000 ms	0 ms	CHORD	
58	Break direction	0 .. 2	Up	CHORD	0=Up, 1=Down, 2=Alternate
59	Voice 1 bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
60	Voice 2 bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
61	Voice 3 bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
62	Voice 4 bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
63	Voice 5 bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
64	Voice 6 bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
65	Voice 7 bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
66	Voice 8 bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
67	Output mode	0 .. 4	All summed 1	SOUND	0=All summed 1, 1=All summed 2, 2=Per gate 12-34, 3=Per gate 12-34 2, 4=Per gate 1-2-3
68	Input mode	0 .. 9	1	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
69	Sustain mode	0 .. 1	Synth	KEYS	0=Synth, 1=Piano
70	MIDI vel curve	0 .. 3	Linear	KEYS	0=Linear, 1=Soft, 2=Hard, 3=Fixed
71	Arp reset input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
72	Gate offset	0 .. 496	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
73	Round robin mode	0 .. 3	Sequential	SOUND	0=Sequential, 1=Random, 2=Random2, 3=Random3
74	Voice 9 detune	-100 .. 100 cents	0 cents	TUNE	

#	Parameter	Range	Default	Page	Values / notes
75	Voice 10 detune	-100 .. 100 cents	0 cents	TUNE	
76	Voice 11 detune	-100 .. 100 cents	0 cents	TUNE	
77	Voice 12 detune	-100 .. 100 cents	0 cents	TUNE	
78	Voice 13 detune	-100 .. 100 cents	0 cents	TUNE	
79	Voice 14 detune	-100 .. 100 cents	0 cents	TUNE	
80	Voice 15 detune	-100 .. 100 cents	0 cents	TUNE	
81	Voice 16 detune	-100 .. 100 cents	0 cents	TUNE	



5 – WAV RECORDER AND AUTO-SAMPLER

Switch to the recorder with the **5 WAV RECORDER** tab. **RECORD** records the plug-in's audio input to *!RECORD/RECnnnnn.WAV* (48 or 96 kHz, 16 or 24-bit, mono or stereo); **PLAY** plays the recording selected in the left list. The display shows level meters with peak hold and clip indicators.

AUTO-SAMPLING AN INSTRUMENT

1. **Connect it.** In the standalone app open *Options -> Audio/MIDI settings*: pick the MIDI output that plays your synth and the audio input it comes back on. (In a DAW, route the plug-in's MIDI out to the synth and the synth's audio into the plug-in - not every DAW allows MIDI out of an instrument, so the app is easiest.) Modular synths can use the CV/gate on Out 3/4.
2. **Plan it** on the AUTO-SAMPLER page: start and end note, note step, number of velocity layers (spread between AS Min and AS Max velocity), round robins, note length (AS Coarse seconds + AS Fine ms) and AS Gap (time left for the release).
3. **Press TEST.** A note plays and the status line shows the measured latency in samples. Set *AS Latency* a little below it so each sample starts right at the attack.
4. **Press AUTO.** Every note × velocity × round robin is played and recorded into a new *AUTOnnnn* folder, already named for SD Multisample (e.g. 060_C4_V1_RR1.wav). Press AUTO again to stop early.
5. **Play it.** Click the folder in the right-hand *Auto-sampled* list and the plug-in switches to SD Multisample with it loaded.

PARAMETERS

#	Parameter	Range	Default	Page	Values / notes
1	Attenuverter 1	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
2	Attenuverter 2	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
3	Attenuverter 3	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
4	Attenuverter 4	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
5	Attenuverter 5	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
6	Attenuverter 6	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
7	Input gain	0 .. 24 dB	0 dB	RECORD	
8	Sample rate	0 .. 1	48kHz	RECORD	0=48kHz, 1=96kHz
9	Bit depth	0 .. 1	16 bit	RECORD	0=16 bit, 1=24 bit
10	Channels	1 .. 6	2	RECORD	
11	Rec CV Mode	0 .. 1	Trigger	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
12	Play CV Mode	0 .. 1	Trigger	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
13	Play As Loop	0 .. 1	One-shot	RECORD	0=One-shot, 1=Loop
14	Outputs 1/2	0 .. 2	Thru/Play	RECORD	0=Thru, 1=Thru/Play, 2=Play
15	Outputs 3/4	0 .. 2	Thru	RECORD	0=Thru, 1=Thru/Play, 2=Play
16	AS Enable	0 .. 1	Off	AUTO-SAMPLER	0=Off, 1=On
17	AS Start Note	0 .. 127	21	AUTO-SAMPLER	
18	AS End Note	0 .. 127	108	AUTO-SAMPLER	
19	AS Note Step	1 .. 127	1	AUTO-SAMPLER	
20	AS Vel Sw	1 .. 9	1	AUTO-SAMPLER	
21	AS RRobins	1 .. 9	1	AUTO-SAMPLER	
22	AS Coarse	0 .. 60 s	1 s	AUTO-SAMPLER	
23	AS Fine	-1000 .. 1000 ms	0 ms	AUTO-SAMPLER	
24	AS Gap	1 .. 600	1	AUTO-SAMPLER	
25	AS Latency	0 .. 960	0	AUTO-SAMPLER	
26	Thru routing 1	0 .. 6	1	RECORD	0=None, 1=1, 2=2, 3=3, 4=4, 5=1&2, 6=3&4
27	Thru routing 2	0 .. 6	2	RECORD	0=None, 1=1, 2=2, 3=3, 4=4, 5=1&2, 6=3&4
28	Thru routing 3	0 .. 6	None	display	0=None, 1=1, 2=2, 3=3, 4=4, 5=1&2, 6=3&4
29	Thru routing 4	0 .. 6	None	display	0=None, 1=1, 2=2, 3=3, 4=4, 5=1&2, 6=3&4
30	Thru routing 5	0 .. 6	None	display	0=None, 1=1, 2=2, 3=3, 4=4, 5=1&2, 6=3&4
31	Thru routing 6	0 .. 6	None	display	0=None, 1=1, 2=2, 3=3, 4=4, 5=1&2, 6=3&4
32	AS Min velocity	0 .. 127	25	AUTO-SAMPLER	
33	AS Max velocity	0 .. 127	127	AUTO-SAMPLER	
34	Playback file	0 .. 9999	0	list	
35	Record enable	0 .. 1	On	display	0=Locked, 1=On

#	Parameter	Range	Default	Page	Values / notes
36	Playback enable	0 .. 1	On	display	0=Locked, 1=On
37	Playback gain	-40 .. 6 dB	0 dB	RECORD	
38	AS Preview note	0 .. 127	48	AUTO-SAMPLER	

EX POLY WAVETABLE



A 16-voice wavetable synthesizer. Pick a wavetable in the left list; the right list shows its waves - click one to jump the *Wave offset* to it. The position can also follow velocity, pitch, both envelopes and the LFO, and *Wave spread* gives every voice a different position. Each voice has two envelopes (with curve shapes; envelope 2's sustain can be negative), a state-variable filter (low/band/high pass) and an LFO; unison stacks up to 8 of the 16 voices per note.

- **OSC** wave offset/spread, unison, voices, gain · **ENV 1**, **ENV 2** · **FILTER** type, cutoff (as a MIDI note), resonance and modulation · **MOD** routings and LFO speed/retrigger/spread · **CHORD** · **FX** Juno-style chorus I/II, delay, output mode (*Per voice* sends voices to Out 1-4) · **KEYS** tuning, sustain, trigger mode (gated, drone, triggered), Scala.
- Wavetables are band-limited (mip-mapped) so high notes stay clean.

YOUR OWN WAVETABLES AND WAVEEDIT

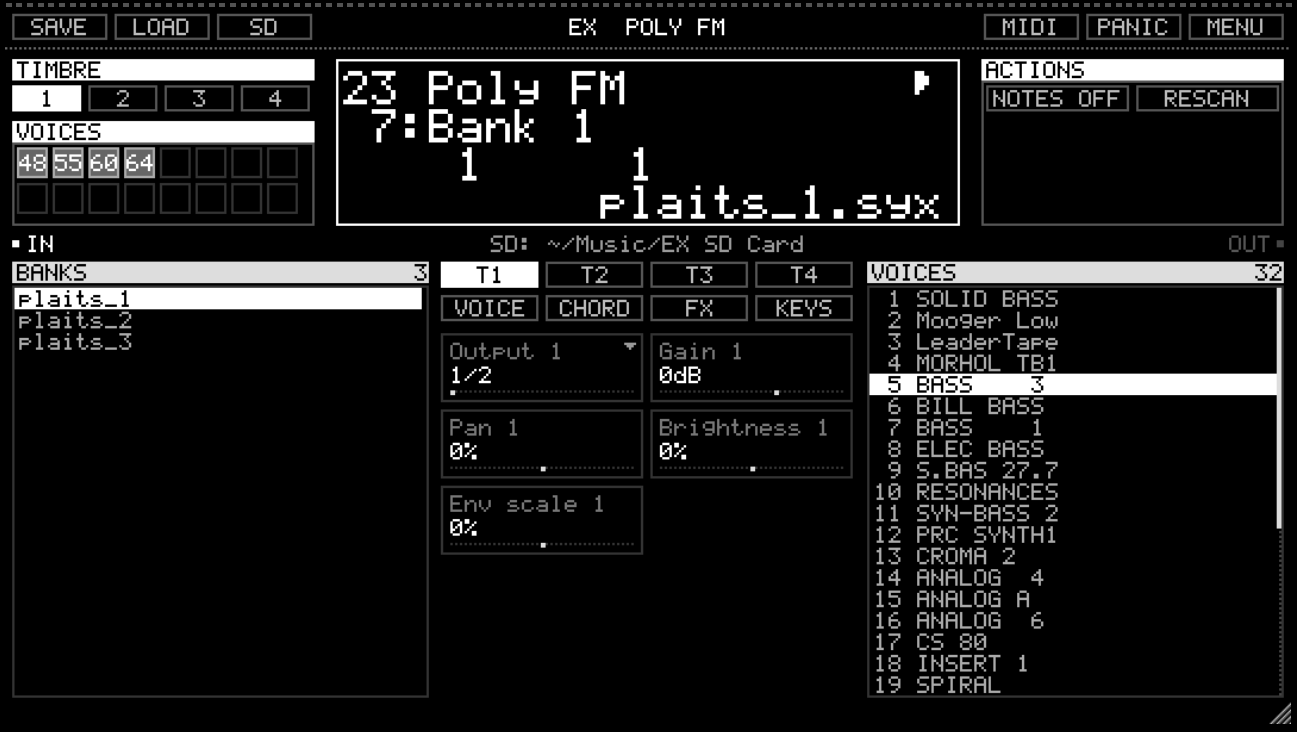
- **Drag and drop** WAV files or folders onto the window, or press **IMPORT**.
- **WAVEEDIT** imports every bank saved by the WaveEdit app in *Documents/WaveEdit/banks*; press it again after editing to update them.
- WaveEdit banks (64 waves × 256 samples) and disting tables (600-sample cycles) are recognised automatically; a set of single-cycle files becomes one table. Each import becomes a folder of single-cycle WAVs in *wavetables/*, so it also works on a real disting EX.

PARAMETERS

#	Parameter	Range	Default	Page	Values / notes
1	Attenuverter 1	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
2	Attenuverter 2	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
3	Attenuverter 3	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
4	Attenuverter 4	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
5	Attenuverter 5	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
6	Attenuverter 6	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
7	Wavetable	0 .. 999	0	list	
8	Wave offset	-100 .. 100	0	OSC	
9	Wave spread	-100 .. 100	0	OSC	
10	Coarse tune	-60 .. 60	0	KEYS	
11	Fine tune	-100 .. 100 cents	0 cents	KEYS	
12	Attack time	0 .. 127	20	ENV 1	
13	Decay time	0 .. 127	60	ENV 1	
14	Sustain level	0 .. 127	80	ENV 1	
15	Release time	0 .. 127	60	ENV 1	
16	Attack shape	0 .. 127	64	ENV 1	
17	Decay shape	0 .. 127	64	ENV 1	
18	Attack time 2	0 .. 127	60	ENV 2	
19	Decay time 2	0 .. 127	70	ENV 2	
20	Sustain level 2	-127 .. 127	64	ENV 2	
21	Release time 2	0 .. 127	50	ENV 2	
22	Attack shape 2	0 .. 127	64	ENV 2	
23	Decay shape 2	0 .. 127	64	ENV 2	
24	Filter type	0 .. 3	Off	FILTER	0=Off, 1=Lowpass, 2=Bandpass, 3=Highpass
25	Filter freq	0 .. 127	64	FILTER	
26	Filter Q	0 .. 100	50	FILTER	
27	Veloc -> volume	0 .. 100 %	100 %	MOD	
28	Veloc -> wave	-100 .. 100 %	0 %	MOD	
29	Veloc -> filter	-127 .. 127	0	FILTER	
30	Pitch -> wave	-100 .. 100 %	0 %	MOD	
31	Pitch -> filter	-100 .. 100 %	0 %	FILTER	
32	Env -> wave	-100 .. 100 %	0 %	MOD	
33	Env -> filter	-127 .. 127	0	FILTER	
34	Env 2 -> wave	-100 .. 100 %	0 %	MOD	
35	Env 2 -> filter	-127 .. 127	0	FILTER	
36	Env 2 -> pitch	-120 .. 120	0	MOD	
37	LFO -> wave	-100 .. 100 %	0 %	MOD	
38	LFO -> filter	-127 .. 127	0	FILTER	
39	LFO -> pitch	-120 .. 120	0	MOD	
40	LFO speed	-100 .. 100	90	MOD	
41	Gain	-40 .. 24 dB	0 dB	OSC	
42	Sustain	0 .. 1	Off	KEYS	0=Off, 1=On
43	Bend range	0 .. 48	2	KEYS	

#	Parameter	Range	Default	Page	Values / notes
44	Chord enable	0 .. 7	Off	CHORD	0=Off, 1=On, 2=Gate 1, 3=Gate 2, 4=Gate 3, 5=Gates 1+2, 6=Gates 1+3, 7=Gates 2+3
45	Chord key	-12 .. 12	0	CHORD	
46	Chord scale	0 .. 7	Major	CHORD	0=Major, 1=Natural Minor, 2=Dominant, 3=Fully Diminished, 4=Dominant Dim, 5=Augmented, 6=Whole Tone, 7=Chromatic
47	Chord shape	0 .. 13	None	CHORD	0=None, 1=Octave, 2=Two Octaves, 3=Root/Fifth, 4=Root/Fifth + 8ve, 5=Triad, 6=Triad + 8ve, 7=Sus4, 8=Sus4 + 8ve, 9=Sixth, 10=Sixth + 8ve, 11=Seventh, 12=Seventh + 8ve, 13=Ninth
48	Chord inversion	0 .. 3	0	CHORD	
49	Arpeggio 1 mode	0 .. 14	Off	display	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
50	Arpeggio 2 mode	0 .. 14	Off	display	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
51	Arpeggio 3 mode	0 .. 14	Off	CHORD	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
52	Arpeggio 1 range	1 .. 3	1	display	
53	Arpeggio 2 range	1 .. 3	1	display	
54	Arpeggio 3 range	1 .. 3	1	CHORD	
55	Scala/MTS	-16 .. 99	0	KEYS	
56	Scala KBM	0 .. 99	0	KEYS	
57	Chorus mode	0 .. 2	Off	FX	0=Off, 1=I, 2=II
58	Delay mode	0 .. 2	Off	FX	0=Off, 1=Stereo, 2=Ping-pong
59	Delay level	-40 .. 0 dB	-3 dB	FX	
60	Delay time	1 .. 10000 ms	500 ms	FX	
61	Delay feedback	0 .. 100 %	50 %	FX	
62	Unison	1 .. 8	1	OSC	
63	Unison detune	0 .. 100 cents	10 cents	OSC	
64	Output spread	-100 .. 100 %	0 %	FX	
65	Output mode	0 .. 3	Spread voice	FX	0=Spread voice, 1=Spread voice 2, 2=Spread pitch, 3=Per voice
66	Input mode	0 .. 18	1	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
67	Sustain mode	0 .. 1	Synth	KEYS	0=Synth, 1=Piano
68	MIDI vel curve	0 .. 3	Linear	KEYS	0=Linear, 1=Soft, 2=Hard, 3=Fixed
69	Wave input	0 .. 6	5	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
70	Pitch bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
71	Arp reset input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
72	Max voices	1 .. 16	16	OSC	
73	LFO retrigger	0 .. 2	Poly	MOD	0=Poly, 1=Mono, 2=Off
74	LFO spread	0 .. 90 deg	0 deg	MOD	
75	Gate offset	0 .. 496	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
76	Trigger mode	0 .. 2	Gated	KEYS	0=Gated, 1=Drone, 2=Triggered
77	Break time	0 .. 1000 ms	0 ms	CHORD	
78	Break direction	0 .. 2	Up	CHORD	0=Up, 1=Down, 2=Alternate

EX POLY FM



A 16-voice, 4-timbre FM synthesizer using the six-operator engine of Mutable Instruments Plaits - the same engine the disting EX uses. Banks (left list) are DX7 32-voice SysEx files from the card's *FMSYX* folder; the right list shows the bank's voices by name. The TIMBRE buttons choose which timbre the lists set; timbres 2-4 can follow timbre 1.

- **T1 … T4**: output, gain, pan, *Brightness* (modulator depth) and *Env scale* (envelope times) per timbre.
- **VOICE**: max voices, unison, master gain, velocity -> brightness, voice transpose and output mode. **CHORD**, **FX** (chorus, delay), **KEYS**.
- Parameters 54-64 are not listed in the module's manual; EX-GF uses them for the voice and effect extras.

EDITING WITH A DX7 EDITOR

Point a DX7 editor/librarian (Dexed, Patch Base, MIDI Quest …) at the plug-in. Single-voice dumps, parameter edits and dump requests apply to the timbre chosen by *SysEx target*; the display then shows **VOICE EDITED**. From MENU -> Poly FM: *Write voice* stores the edit in the bank (in memory), *Write bank to SD* saves it, *Abandon edit* reverts, and *Send* dumps the voice or bank back to the editor. *Write built-in to SD* creates FMSYX with the built-in banks.

PARAMETERS

#	Parameter	Range	Default	Page	Values / notes
1	Attenuverter 1	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
2	Attenuverter 2	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
3	Attenuverter 3	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
4	Attenuverter 4	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
5	Attenuverter 5	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
6	Attenuverter 6	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
7	Bank 1	1 .. 999	1	list	
8	Voice 1	1 .. 32	1	list	
9	Output 1	0 .. 7	1/2	T1	0=1/2, 1=3/4, 2=1, 3=2, 4=3, 5=4, 6=1/2+3, 7=1/2+4
10	Gain 1	-40 .. 24 dB	0 dB	T1	
11	Pan 1	-100 .. 100 %	0 %	T1	
12	Brightness 1	-100 .. 100 %	0 %	T1	
13	Env scale 1	-100 .. 100 %	0 %	T1	
14	Bank 2	0 .. 999	0	list	
15	Voice 2	0 .. 32	0	list	
16	Output 2	0 .. 7	1/2	T2	0=1/2, 1=3/4, 2=1, 3=2, 4=3, 5=4, 6=1/2+3, 7=1/2+4
17	Gain 2	-40 .. 24 dB	0 dB	T2	
18	Pan 2	-100 .. 100 %	0 %	T2	
19	Brightness 2	-100 .. 100 %	0 %	T2	
20	Env scale 2	-100 .. 100 %	0 %	T2	
21	Bank 3	0 .. 999	0	list	
22	Voice 3	0 .. 32	0	list	
23	Output 3	0 .. 7	1/2	T3	0=1/2, 1=3/4, 2=1, 3=2, 4=3, 5=4, 6=1/2+3, 7=1/2+4
24	Gain 3	-40 .. 24 dB	0 dB	T3	
25	Pan 3	-100 .. 100 %	0 %	T3	
26	Brightness 3	-100 .. 100 %	0 %	T3	
27	Env scale 3	-100 .. 100 %	0 %	T3	
28	Bank 4	0 .. 999	0	list	
29	Voice 4	0 .. 32	0	list	
30	Output 4	0 .. 7	1/2	T4	0=1/2, 1=3/4, 2=1, 3=2, 4=3, 5=4, 6=1/2+3, 7=1/2+4
31	Gain 4	-40 .. 24 dB	0 dB	T4	
32	Pan 4	-100 .. 100 %	0 %	T4	
33	Brightness 4	-100 .. 100 %	0 %	T4	
34	Env scale 4	-100 .. 100 %	0 %	T4	
35	Coarse tune	-60 .. 60 ST	0 ST	KEYS	
36	Fine tune	-100 .. 100 cents	0 cents	KEYS	
37	Sustain	0 .. 1	Off	KEYS	0=Off, 1=On
38	Bend range	0 .. 48	2	KEYS	
39	Chord enable	0 .. 7	Off	CHORD	0=Off, 1=On, 2=Gate 1, 3=Gate 2, 4=Gate 3, 5=Gates 1+2, 6=Gates 1+3, 7=Gates 2+3
40	Chord key	-12 .. 12	0	CHORD	

#	Parameter	Range	Default	Page	Values / notes
41	Chord scale	0 .. 7	Major	CHORD	0=Major, 1=Natural Minor, 2=Dominant, 3=Fully Diminished, 4=Dominant Dim, 5=Augmented, 6=Whole Tone, 7=Chromatic
42	Chord shape	0 .. 13	None	CHORD	0=None, 1=Octave, 2=Two Octaves, 3=Root/Fifth, 4=Root/Fifth + 8ve, 5=Triad, 6=Triad + 8ve, 7=Sus4, 8=Sus4 + 8ve, 9=Sixth, 10=Sixth + 8ve, 11=Seventh, 12=Seventh + 8ve, 13=Ninth
43	Chord inversion	0 .. 3	0	CHORD	
44	Arpeggio 1 mode	0 .. 14	Off	display	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
45	Arpeggio 2 mode	0 .. 14	Off	display	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
46	Arpeggio 3 mode	0 .. 14	Off	CHORD	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
47	Arpeggio 1 range	1 .. 3	1	display	
48	Arpeggio 2 range	1 .. 3	1	display	
49	Arpeggio 3 range	1 .. 3	1	CHORD	
50	Scala/MTS	-16 .. 99	0	KEYS	
51	Scala KBM	0 .. 99	0	KEYS	
52	Output mode	0 .. 2	Spread voice	VOICE	0=Spread voice, 1=Spread voice 2, 2=Spread pitch
53	Output spread	-100 .. 100 %	0 %	VOICE	
54	Max voices	1 .. 16	16	VOICE	
55	Unison	1 .. 8	1	VOICE	
56	Unison detune	0 .. 100 cents	10 cents	VOICE	
57	Chorus mode	0 .. 2	Off	FX	0=Off, 1=I, 2=II
58	Delay mode	0 .. 2	Off	FX	0=Off, 1=Stereo, 2=Ping-pong
59	Delay level	-40 .. 0 dB	-3 dB	FX	
60	Delay time	1 .. 10000 ms	500 ms	FX	
61	Delay feedback	0 .. 100 %	50 %	FX	
62	Gain	-40 .. 24 dB	0 dB	VOICE	
63	Veloc -> bright	-100 .. 100 %	0 %	VOICE	
64	Voice transpose	0 .. 1	Use	VOICE	0=Ignore, 1=Use
65	Input mode	0 .. 9	1	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
66	Sustain mode	0 .. 1	Synth	KEYS	0=Synth, 1=Piano
67	MIDI vel curve	0 .. 3	Linear	KEYS	0=Linear, 1=Soft, 2=Hard, 3=Fixed
68	Pitch bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
69	Arp reset input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
70	Gate offset	0 .. 496	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
71	Trigger mode	0 .. 2	Gated	KEYS	0=Gated, 1=Drone, 2=Triggered
72	Break time	0 .. 1000 ms	0 ms	CHORD	
73	Break direction	0 .. 2	Up	CHORD	0=Up, 1=Down, 2=Alternate
74	SysEx target	1 .. 4	1	KEYS	

EX POLY EXCITER



A 16-voice synthesizer that excites a resonating string with a short burst of sound - plucked, struck, scraped or bowed. The left list holds exciter tables (*Internal*, or tables from the card's *exciters* folder); the right list shows the exciters in the table. Drum hits make great exciters: a kick for a soft tone, a snare for a sharp, noisy attack.

Mode	Behaviour
Strike	The exciter plays once, for the whole Exciter length.
Scrape	As Strike, but the exciter stops when the key is released.
Strum	The exciter loops while the key is held; its level follows **pressure** (aftertouch or the Pressure parameter) - silent without it.

Internal exciter	Sound
0 Ramp	A decaying linear ramp.
1 Noise burst	White noise at a constant level.
2 Noise ramp	Noise shaped by a decaying ramp.
3 Square	Constant level: strong plucks at the start and end.
4 Silence	For processing external audio only.

- **EXCITER** mode, exciter, length, pressure, *Ext exciter input* (feed audio into the plug-in to excite the strings live).
- **RESONATOR** tone (brightness), damping, extra damping on release, low-pass filter. **VELO** velocity/pitch/pressure modulation. **ENV/LFO** modulate pitch and exciter choice. **CHORD**, **FX**, **KEYS**.

PARAMETERS

#	Parameter	Range	Default	Page	Values / notes
1	Attenuverter 1	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
2	Attenuverter 2	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
3	Attenuverter 3	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
4	Attenuverter 4	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
5	Attenuverter 5	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
6	Attenuverter 6	-200 .. 200 %	100 %	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
7	Mode	0 .. 2	Strike	EXCITER	0=Strike, 1=Scrape, 2=Strum
8	Table	-1 .. 999	-1	list	
9	Exciter	0 .. 127	0	EXCITER	
10	Exciter length	1 .. 597 ms	100 ms	EXCITER	
11	Tone	0 .. 127	50	RESONATOR	
12	Damp	0 .. 100	4	RESONATOR	
13	Release damp	0 .. 100	0	RESONATOR	
14	Pressure	0 .. 127	0	EXCITER	
15	Veloc -> strike	0 .. 100 %	100 %	VELO	
16	Veloc -> tone	-100 .. 100 %	0 %	VELO	
17	Veloc -> damp	-100 .. 100 %	0 %	VELO	
18	Veloc -> exciter	-127 .. 127	0	VELO	
19	Pitch -> damp	-100 .. 100 %	0 %	VELO	
20	Pitch -> exciter	-100 .. 100 %	0 %	VELO	
21	Pitch -> LPF	-100 .. 100 %	0 %	RESONATOR	
22	Press -> exciter	-127 .. 127	0	VELO	
23	Env -> pitch	-120 .. 120	0	ENV/LFO	
24	Env -> exciter	-127 .. 127	0	ENV/LFO	
25	LFO -> pitch	-200 .. 200 cents	0 cents	ENV/LFO	
26	LFO -> exciter	-127 .. 127	0	ENV/LFO	
27	LFO speed	-100 .. 100	90	ENV/LFO	
28	Attack time	0 .. 127	60	ENV/LFO	
29	Decay time	0 .. 127	70	ENV/LFO	
30	Sustain level	-127 .. 127	0	ENV/LFO	
31	Release time	0 .. 127	50	ENV/LFO	
32	Coarse tune	-60 .. 60	0	KEYS	
33	Fine tune	-100 .. 100 cents	0 cents	KEYS	
34	Gain	-40 .. 24 dB	0 dB	RESONATOR	
35	Sustain	0 .. 1	Off	KEYS	0=Off, 1=On
36	Bend range	0 .. 48	2	KEYS	
37	Chord enable	0 .. 7	Off	CHORD	0=Off, 1=On, 2=Gate 1, 3=Gate 2, 4=Gate 3, 5=Gates 1+2, 6=Gates 1+3, 7=Gates 2+3
38	Chord key	-12 .. 12	0	CHORD	
39	Chord scale	0 .. 7	Major	CHORD	0=Major, 1=Natural Minor, 2=Dominant, 3=Fully Diminished, 4=Dominant Dim, 5=Augmented, 6=Whole Tone, 7=Chromatic

#	Parameter	Range	Default	Page	Values / notes
40	Chord shape	0 .. 13	None	CHORD	0=None, 1=Octave, 2=Two Octaves, 3=Root/Fifth, 4=Root/Fifth + 8ve, 5=Triad, 6=Triad + 8ve, 7=Sus4, 8=Sus4 + 8ve, 9=Sixth, 10=Sixth + 8ve, 11=Seventh, 12=Seventh + 8ve, 13=Ninth
41	Chord inversion	0 .. 3	0	CHORD	
42	Arpeggio 1 mode	0 .. 14	Off	display	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
43	Arpeggio 2 mode	0 .. 14	Off	display	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
44	Arpeggio 3 mode	0 .. 14	Off	CHORD	0=Off, 1=Up, 2=Down, 3=Alt, 4=Alt2, 5=Up -8ve, 6=Down -8ve, 7=Alt -8ve, 8=Alt2 -8ve, 9=Step Up, 10=Step Down, 11=Random, 12=Random2, 13=Random3, 14=As Played
45	Arpeggio 1 range	1 .. 3	1	display	
46	Arpeggio 2 range	1 .. 3	1	display	
47	Arpeggio 3 range	1 .. 3	1	CHORD	
48	Scala/MTS	-16 .. 99	0	KEYS	
49	Scala KBM	0 .. 99	0	KEYS	
50	Low pass filter	0 .. 127	126	RESONATOR	
51	Chorus mode	0 .. 2	Off	FX	0=Off, 1=I, 2=II
52	Delay mode	0 .. 2	Off	FX	0=Off, 1=Stereo, 2=Ping-pong
53	Delay level	-40 .. 0 dB	-3 dB	FX	
54	Delay time	1 .. 10000 ms	500 ms	FX	
55	Delay feedback	0 .. 100 %	50 %	FX	
56	Unison	1 .. 8	1	FX	
57	Unison detune	0 .. 100 cents	10 cents	FX	
58	Output spread	-100 .. 100 %	0 %	FX	
59	Break time	0 .. 1000 ms	0 ms	CHORD	
60	Break direction	0 .. 2	Up	CHORD	0=Up, 1=Down, 2=Alternate
61	Output mode	0 .. 3	Spread voice	FX	0=Spread voice, 1=Spread voice 2, 2=Spread pitch, 3=Per voice
62	Input mode	0 .. 18	1	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
63	Sustain mode	0 .. 1	Synth	KEYS	0=Synth, 1=Piano
64	MIDI vel curve	0 .. 3	Linear	KEYS	0=Linear, 1=Soft, 2=Hard, 3=Fixed
65	Ext exciter input	0 .. 6	None	EXCITER	0=None, 1=In L, 2=In R, 3=In L+R, 4=In L+R, 5=In L+R, 6=In L+R
66	Pitch bend input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
67	Arp reset input	0 .. 6	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*
68	Max voices	1 .. 16	16	KEYS	
69	LFO retrigger	0 .. 1	Poly	ENV/LFO	0=Poly, 1=Off
70	LFO spread	0 .. 90 deg	0 deg	ENV/LFO	
71	Gate offset	0 .. 496	0	-	*CV/hardware only - kept for numbering, no effect in the plug-in*

TROUBLESHOOTING

Problem	Fix
macOS says the app is damaged or from an unidentified developer; the DAW skips the plug-ins	The builds are not notarised. Run the `xattr -dr com.apple.quarantine …` command from the install chapter, then rescan plug-ins.
No sound from EX Multisample	Check the SD folder has sample folders (SD -> Show in Finder) and that a folder is selected in the left list.
Notes hang	Press PANIC (or MIDI -> All notes off).
The DAW does not see Out 3/4	Enable the plug-in's second output bus in the DAW's routing/IO settings.
Auto-sampled files start late or early	Run TEST and set AS Latency just below the measured value.
Large folders take a moment to load	Samples load into memory in the background; very large libraries (hundreds of MB) take a few seconds.
A wavetable change does nothing	On the display, Wavetable needs confirming (PUSH V): press Esc, or choose the table in the list instead.

CREDITS AND LICENCE

- EX-GF is open source under the **GNU AGPL-3.0**; source code at github.com/femifleming/ex-gf.
- Built with JUCE 9 (AGPLv3). Poly FM uses the FM engine of Mutable Instruments Plaits by Émilie Gillet (MIT).
- Behaviour follows the disting EX user manual by Expert Sleepers. Not affiliated with Expert Sleepers.
- Sample library: VSCO-2 Community Edition, Karoryfer (Emily Guitar, Growlybass, Hadzi-Fia), Virtuosity Drums, AKWF wavetables, Greg Sullivan CP80 (CC BY 3.0) and generated content - see CREDITS.txt in the library for details.